MULTI – LANGUAGE SPEECH RECOGNITION

SPEECH AND LANGUAGE PROCESSING

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ABSTRACT

Automatic Speech Recognition (ASR) for multilingual and code-switched speech remains a challenging problem, particularly in India where speakers frequently switch between languages such as Hindi and English within the same utterance. This project presents Indian Code-Switched ASR & Text Analysis, a full-stack research application designed to study multilingual code-switched speech and text, beginning with Hindi-English (Hinglish) and providing a foundation for extending the system to other Indian languages. The system integrates multiple ASR backends, including OpenAI Whisper, the Oriserve Whisper-Hindi2Hinglish-Swift fine-tuned model, and AI4Bharat's IndicConformer, allowing their performance to be compared within a common pipeline.

The application provides two modes of analysis: a live speech pipeline and a direct text-analysis interface. For speech input, the system performs audio preprocessing, multilingual ASR, token-level language identification, code-switch detection, post-processing, translation, and evaluation. The language identification component uses script detection, a Romanized-Hindi lexicon, English word frequencies, and contextual information to classify tokens, with the method explicitly treated as approximate. The evaluation framework supports Word Error Rate (WER), Character Error Rate (CER), token-level language identification accuracy, and code-switch detection precision, recall, and F1-score. The system also includes reproducible experiment execution using speaker-independent dataset splits and supports datasets such as MUCS and VITB-HEBiC.

Initial results from an 8-utterance smoke test indicate that the fine-tuned Hinglish Swift model performs better than the zero-shot Whisper Small baseline while also requiring substantially less inference time. These results suggest that domain-specific fine-tuning can provide meaningful improvements for code-switched speech recognition while maintaining greater computational efficiency with a considerably smaller model. The project emphasizes reproducible experimentation and measurement honesty, providing a practical framework for studying and improving multilingual ASR for Indian code-switched speech.

LITERATURE REVIEW

1. ***Robust Speech Recognition via Large-Scale Weak Supervision* — Alec Radford et al. (2022)**

Radford et al. introduced Whisper, a large-scale multilingual speech recognition system trained on approximately 680,000 hours of weakly supervised audio data collected from the web. The study demonstrated that training on diverse multilingual and multitask speech data enables strong generalization across languages, accents, and different speech conditions without requiring task-specific fine-tuning. Whisper achieved competitive performance across several speech recognition benchmarks and showed improved robustness compared with many conventional supervised ASR systems. The availability of pretrained models also made Whisper practical for real-world applications. This work is particularly relevant to the proposed system because Whisper is used as the primary ASR model for generating multilingual speech transcriptions and serves as the foundation for subsequent language identification and code-switching analysis.

2. ***Phone Merging for Code-Switched Speech Recognition* — Sivasankaran et al. (2018)**

Sivasankaran et al. investigated the challenges of Automatic Speech Recognition for Hindi-English code-switched speech, focusing on the interaction between the phonetic characteristics of the two languages. The authors compared conventional acoustic models using separate phoneme sets for Hindi and English with approaches that merged phones shared across both languages. Their experiments showed that sharing phonetic information between the languages can improve recognition performance, demonstrating that code-switched speech benefits from models that account for interactions between languages rather than treating each language independently. This work is relevant to the proposed system because it highlights the specific difficulties associated with Hindi-English code-switching and supports the need for language-aware processing in multilingual speech recognition.

3. ***MuST-C: A Multilingual Speech Translation Corpus* — Mauro Antonio Di Gangi et al. (2019)**

Di Gangi et al. introduced MuST-C, a large-scale multilingual speech translation corpus developed to address the limited availability of high-quality, publicly available speech resources for multilingual research. The corpus consists of hundreds of hours of English TED Talks covering multiple target languages, with each speech segment accompanied by manually produced transcriptions and translations. The dataset was created by aligning speech recordings with their corresponding textual content, providing researchers with both acoustic and linguistic information in a consistent format. The authors also established baseline results for automatic speech translation, demonstrating that the corpus could effectively support the development and evaluation of end-to-end multilingual speech processing systems. One of the major contributions of the work is its scale and diversity, which make it possible to investigate multilingual speech processing across several languages using a common dataset and evaluation framework. Although the present project focuses primarily on speech recognition and code-switched speech rather than speech translation, MuST-C is relevant because it demonstrates the importance of large, diverse, accurately transcribed, and well-aligned speech datasets for developing reliable multilingual ASR systems. The work also highlights how the availability of standardized multilingual speech resources can accelerate research and provide consistent benchmarks for comparing different speech recognition and processing approaches.

4. ***End-to-End ASR for Code-switched Hindi-English Speech* — Brij Mohan Lal Srivastava, Basil Abraham, Sunayana Sitaram, Rupesh K. Mehta, and Preethi Jyothi (2019)**

Srivastava et al. investigated the use of end-to-end Automatic Speech Recognition models for Hindi-English code-switched speech under low-resource conditions. Unlike many existing studies that rely on hundreds or thousands of hours of training data, the authors worked with less than 50 hours of code-switched speech, reflecting the limited availability of suitable datasets for Indian languages. The study explored Connectionist Temporal Classification (CTC) and attention-based architectures and introduced multi-task learning and corpus balancing to address the challenges caused by limited data and uneven grapheme distributions. The results showed that although limited training data negatively affects end-to-end ASR performance, multi-task learning and data balancing can provide meaningful improvements. This work is highly relevant to the proposed system because it demonstrates the difficulties of Hindi-English code-switched ASR and highlights the importance of handling limited and imbalanced multilingual speech data.

5. ***Code Switched and Code Mixed Speech Recognition for Indic Languages* — Harveen Singh Chadha, Priyanshi Shah, Ankur Dhuriya, Neeraj Chhimwal, Anirudh Gupta, and Vivek Raghavan (2022)**

Chadha et al. studied multilingual and code-switched speech recognition for Indic languages, focusing on the difficulty of building effective ASR systems when acoustic and lexical characteristics differ across languages. The authors compared multilingual end-to-end ASR models with monolingual ASR systems conditioned using Language Identification (LID). Their approach used the language information obtained from a multilingual model to guide the selection and combination of monolingual models, resulting in significant improvements in recognition performance. The study reported substantial reductions in Word Error Rate across multiple Indic language combinations, including Hindi-English and Bengali-English code-switching. The work demonstrates that language identification can play an important role in improving multilingual ASR rather than treating language identification and speech recognition as completely separate tasks. This directly relates to the proposed project, where ASR is combined with language identification and code-switch detection to analyse multilingual speech.

6. ***An Improved Transformer Transducer Architecture for Hindi-English Code Switched Speech Recognition* — Ansen Antony, Sumanth Reddy Kota, Akhilesh Lade, Spoorthy V., and Shashidhar G. Koolagudi (2022)**

Antony et al. proposed an improved Transformer-Transducer architecture specifically for Hindi-English code-switched speech recognition. The study addresses a major limitation of end-to-end ASR systems: their dependence on large amounts of labelled code-switched speech, which is difficult to obtain. To overcome this limitation, the authors leveraged the availability of larger monolingual speech datasets by pre-training language-specific components of the Transformer architecture before applying them to the code-switched recognition task. The proposed approach achieved a Word Error Rate of 29.63% and a Transliterated Word Error Rate of 27.42%, improving upon the existing state of the art at the time. The paper is particularly relevant to this project because it demonstrates how pretrained or language-specific knowledge can be used to improve Hindi-English code-switched ASR when dedicated code-switched training data is limited.

7. ***MUCS 2021: Multilingual and Code-Switching ASR Challenges for Low Resource Indian Languages*** **— Anuj Diwan et al. (2021)**

Diwan et al. introduced the MUCS 2021 challenge to encourage research in multilingual and code-switched Automatic Speech Recognition for low-resource Indian languages. The challenge covered seven Indian languages, including Hindi, Marathi, Odia, Tamil, Telugu, Gujarati, and Bengali, along with Hindi-English and Bengali-English code-switched speech. The authors provided approximately 600 hours of transcribed speech data and established baseline systems for both multilingual and code-switching tasks. The study highlights the difficulties caused by differences in acoustic and linguistic characteristics between Indian languages, as well as the limited availability of labelled speech data. The work is highly relevant to the proposed project because it establishes an important benchmark for Indian multilingual ASR and specifically identifies Hindi-English code-switching as an important research problem.

8. ***Dual Script E2E Framework for Multilingual and Code-Switching ASR*** **— Mari Ganesh Kumar et al. (2021)**

Kumar et al. proposed a dual-script end-to-end framework for multilingual and code-switched ASR, addressing the additional difficulty created by the different writing systems used by Indian languages. The authors introduced a common phoneme-level label representation that allows multiple languages to be represented using a shared output space. Two end-to-end systems were investigated: one trained using the common label representation followed by a data-driven backend for recovering the native script, and another that simultaneously uses the common representation and native-language characters during training. This approach attempts to exploit similarities between Indian languages while still preserving their individual scripts. The study is relevant to the proposed work because multilingual Indian speech often involves different linguistic and writing systems, and it demonstrates how a shared representation can help build a single ASR system capable of handling multilingual and code-switched speech.

9. ***The CSTR System for Multilingual and Code-Switching ASR Challenges for Low Resource Indian Languages*** **— Ondřej Klejch, Electra Wallington, and Peter Bell (2021)**

Klejch et al. presented the CSTR system developed for the MUCS 2021 multilingual and code-switching ASR challenge. For the multilingual task, the authors trained a multilingual CNN-TDNN acoustic model covering several Indian languages and subsequently fine-tuned it using language-specific data. They also incorporated external text from Common Crawl and YouTube data to improve acoustic and language modelling. Confidence-based language identification was used to help determine the appropriate language during recognition, resulting in an average Word Error Rate of 18.1% and a substantial improvement over the challenge baseline. For the code-switching task, the system incorporated multilingual language modelling and transliteration techniques to handle English terms occurring within Indian-language speech. This work is particularly relevant to the proposed project because it demonstrates the practical value of combining multilingual ASR with language identification, external data, and transliteration when dealing with low-resource code-switched Indian speech.

10. ***SRI-B End-to-End System for Multilingual and Code-Switching ASR Challenges for Low Resource Indian Languages*** **— Hardik Sailor et al. (2021)**

Sailor et al. developed an end-to-end multilingual ASR system for the MUCS 2021 challenge using a Transformer architecture trained with both Connectionist Temporal Classification and Cross-Entropy objectives. The authors observed that simply pooling speech data from several languages can improve generalization but may reduce performance compared with language-specific models. To address this issue, they introduced language-specific embeddings derived from an attention-based language identification model and used these embeddings to condition the multilingual ASR system. The system also employed language-specific adaptation and a hypothesis elimination strategy based on language identification scores to select the most suitable recognition output. Their experiments showed that incorporating explicit language information improved recognition performance over the baseline system. This work is closely connected to the proposed project because it demonstrates how language identification can be integrated with multilingual ASR rather than being treated as an independent post-processing task.

11. ***VITB-HEBiC: A Bilingual Corpus for Evaluating ASR in Diverse Indian Code-Switching Scenarios*** **— N. Jose et al. (2024)**

The authors introduced VITB-HEBiC, a Hindi-English bilingual speech corpus developed specifically for evaluating Automatic Speech Recognition systems under diverse Indian code-switching conditions. The corpus contains approximately 7.5 hours of read speech, 3,590 utterances, and more than 58,000 words collected from speakers representing 27 different states of India. This geographical diversity allows the dataset to capture variations in accents, dialects, and pronunciation that are commonly encountered in real-world Indian speech. The study evaluated several neural ASR architectures, including Whisper, Wav2Vec 2.0, XLSR-53, W2V2-BERT, and IndicWav2Vec-Hindi, establishing useful baseline results for Hindi-English code-switched recognition. The study highlights the shortage of diverse code-switched speech resources and demonstrates the importance of including speakers with different linguistic backgrounds when evaluating multilingual ASR systems. This work is closely related to the proposed project because it focuses specifically on Hindi-English code-switching and provides a practical basis for evaluating modern multilingual ASR models such as Whisper.

12. ***Enhancing Whisper's Accuracy and Speed for Indian Languages through Prompt-Tuning and Tokenization*** **— Kumud Tripathi, Raj Gothi, and Pankaj Wasnik (2024)**

Tripathi et al. investigated methods for improving the performance of Whisper on low-resource Indian languages. Although Whisper performs well across many languages, its performance can be weaker for languages with limited representation in its training data. The authors proposed two approaches: prompt-tuning using language-family information and a modified tokenizer designed to reduce the number of generated tokens. Prompt-tuning provides the model with additional linguistic context, helping it better distinguish and process languages that share similarities, while the tokenizer aims to reduce inference time by producing fewer tokens during transcription. Experiments were conducted across multiple Whisper model sizes, including Small, Medium, and Large, and showed improvements in recognition accuracy through language-aware prompting while simultaneously improving inference efficiency through the proposed tokenization strategy. This paper is particularly relevant to the proposed system because it demonstrates that a pretrained Whisper model can be further adapted to improve its performance on Indian languages without developing an ASR system completely from scratch.

13. ***Analysing Code-Switching Scenarios in India's Diverse Linguistic Landscape Using End-to-End ASR Systems with VITB-HEBiC* — Authors from the VITB-HEBiC study (2024)**

This study further investigates the performance of modern end-to-end ASR models on diverse Indian code-switching scenarios using the VITB-HEBiC corpus. The researchers evaluated several pretrained models, including Whisper-medium, Wav2Vec 2.0, XLSR-53, W2V2-BERT, and IndicWav2Vec-Hindi, to understand how different architectures perform when processing multilingual speech from speakers with varied Indian linguistic backgrounds. Among the evaluated models, Whisper-medium achieved the lowest Word Error Rate of 15.7%, while W2V2-BERT produced the best Character Error Rate of 6.4%. The comparison demonstrates that different ASR architectures can behave differently depending on the evaluation metric and characteristics of the speech data. The study is relevant to this project because it provides evidence for using pretrained multilingual models as practical baselines for Indian code-switched speech and reinforces the importance of evaluating both word-level and character-level recognition performance.

14. ***Advancing ASR for Indian-Accented English: Dataset Creation and Whisper Fine-Tuning* — Authors from the Amrita Vishwa Vidyapeetham study (2025)**

This study focuses on improving Automatic Speech Recognition for Indian-accented English by creating a large and diverse speech dataset and fine-tuning Whisper models on the collected data. The researchers collected approximately 170 hours of speech from 633 speakers across different regions of India, capturing variations in pronunciation, intonation, regional accents, and linguistic backgrounds. The study then fine-tuned different Whisper model sizes and evaluated their performance using Word Error Rate. The fine-tuned Whisper-Tiny, Small, and Medium models achieved WER values of 18.141%, 17.36%, and 15.08%, respectively, demonstrating that adapting a pretrained multilingual model to region-specific speech can substantially improve recognition performance. The findings are relevant to the proposed project because they show a clear path for improving the current pretrained Whisper baseline through fine-tuning on Indian speech data, which can be particularly valuable for handling the pronunciation and vocabulary variations present in multilingual and code-switched Indian speech.

15. ***Hindi-English Code-Switching Speech Corpus*** **— Ganji Sreeram, Kunal Dhawan, and Rohit Sinha (2018)**

Sreeram et al. developed a Hindi-English code-switching speech corpus to address the limited availability of suitable datasets for training and evaluating code-switched Automatic Speech Recognition systems. The corpus was designed to capture natural variations between speakers and recording sessions, including differences in pronunciation, accent, age, and gender. The authors analysed the characteristics and statistics of the collected speech data and subsequently evaluated its usefulness for the ASR task. The corpus was also designed to support other speech-processing applications such as language identification, language modelling, and speech synthesis. The study emphasizes that code-switched speech presents challenges that are not adequately represented in conventional monolingual datasets, making dedicated bilingual resources important for developing robust systems. This work is directly relevant to the proposed project because Hindi-English speech is the primary code-switching scenario being considered, and it demonstrates the importance of suitable bilingual speech data for evaluating ASR and language identification systems.

16. ***Unified Model for Code-Switching Speech Recognition and Language Identification Based on Concatenated Tokenizer*** **— Kunal Dhawan, Dima Rekesh, and Boris Ginsburg (2023)**

Dhawan et al. proposed a unified approach that combines code-switched speech recognition with language identification within a single ASR model. The authors introduced a Concatenated Tokenizer that combines the tokenization systems of individual languages and allows the model to generate language information for each emitted text token. This approach provides a way of determining not only the transcription of an utterance but also the language associated with different portions of the output. The study also proposed a method for creating code-switched training data from existing monolingual speech resources, helping address the limited availability of naturally occurring code-switched datasets. Experiments were conducted on English-Hindi and English-Spanish code-switched speech, where the proposed approach achieved strong ASR and spoken language identification performance. This paper is particularly relevant to the proposed system because token-level language identification is one of its central components, and the work demonstrates how ASR and language identification can be closely integrated rather than handled as completely independent tasks.

17. ***Multilingual Speech Recognition for Low-Resource Indian Languages using Multi-Task Conformer*** **— Krishna D. N. (2021)**

Krishna D. N. proposed a multi-task learning framework based on the Conformer architecture for multilingual speech recognition in low-resource Indian languages. The model uses a shared Conformer encoder together with two parallel Transformer decoders: one for phoneme recognition and another for grapheme prediction. Phoneme recognition is treated as an auxiliary task, allowing the model to learn additional acoustic and linguistic information while simultaneously performing the main grapheme recognition task. The authors employed joint CTC-Attention training and introduced conditional decoding to provide language information to the model before predicting the grapheme sequence. Experiments showed that the proposed dual-decoder Conformer architecture performed better than comparable Transformer-based and single-decoder approaches, while also demonstrating the potential advantages of multilingual modelling over separate monolingual systems. This work is relevant to the proposed project because it focuses specifically on low-resource Indian languages and demonstrates how Conformer-based architectures and explicit language information can improve multilingual ASR performance.

18. ***Towards Building ASR Systems for the Next Billion Users*** **— Tahir Javed et al. (2022)**

Javed et al. presented IndicWav2Vec, a multilingual speech representation model developed specifically for the large number of Indian language speakers who remain underserved by existing speech technologies. The model was pretrained on speech from 40 Indian languages covering multiple language families and was subsequently fine-tuned for ASR in nine languages. The authors also constructed large publicly available speech corpora and evaluated the resulting models on benchmarks including MUCS, MSR, and OpenSLR. Their results demonstrated strong performance across several Indian languages, showing that multilingual self-supervised speech representations can be effectively transferred to low-resource ASR tasks. The work is particularly relevant to the proposed system because it focuses on the Indian multilingual environment and demonstrates the importance of large-scale multilingual pretraining for improving speech recognition in languages with limited labelled resources.

19. ***A State-of-the-Art ASR Model for Indian Languages*** **— Kaushal Bhogale et al. (2024)**

Bhogale et al. developed IndicConformer, a multilingual ASR model designed specifically for Indian languages using the Conformer architecture. The model combines convolutional layers, which capture local acoustic patterns, with Transformer-based attention mechanisms that capture longer-range dependencies in speech. IndicConformer was developed to provide accurate speech-to-text recognition across the 22 officially recognized Indian languages, addressing the lack of strong open-source ASR models covering such a broad range of languages. The model was trained using multiple Indian speech resources and was designed to support both multilingual and language-specific recognition. Its focus on Indian languages makes it especially relevant to this project, where multilingual speech and Hindi-English code-switching are central concerns. IndicConformer also provides an important comparison point against general-purpose multilingual models such as Whisper and can serve as an Indian-language-specific baseline for evaluating ASR performance.

20. ***Language Identification and Analysis of Code-Switched Social Media Text*** **— Deepthi Mave, Suraj Maharjan, and Thamar Solorio (2018)**

Mave et al. investigated language identification in code-switched text, with particular attention to Hindi-English and Spanish-English language combinations. The authors created a Hindi-English code-switched dataset and evaluated different word-level language identification approaches, including Conditional Random Fields and neural network-based models. Their experiments showed that the CRF-based approach achieved better performance than the neural models on the evaluated datasets, demonstrating the difficulty of identifying the language of individual words in code-switched content. The study also examined different measures for characterizing code-switching patterns, providing a broader analysis beyond simple language classification. Although the work focuses on social media text rather than speech, its token-level language identification approach is highly relevant to the proposed system, where language labels are assigned to recognized tokens in order to identify and analyse Hindi-English code-switching.

OUTPUT

The developed system provides a complete pipeline for analysing multilingual and code-switched speech as well as directly entered text. For speech input, the system processes the audio through preprocessing, ASR, token-level language identification, code-switch detection, post-processing, translation, and evaluation stages. The live dashboard displays these stages progressively through the FastAPI backend, along with measured processing times. The system supports three ASR backends: OpenAI Whisper, Whisper-Hindi2Hinglish-Swift, and AI4Bharat IndicConformer, allowing their outputs and performance to be compared within the same framework. The text-analysis interface provides similar language identification, code-switch statistics, and translation functionality without requiring an audio input.

Initial experimental evaluation was performed using an 8-utterance smoke-test dataset on an Apple M1 CPU. The fine-tuned Whisper-Hindi2Hinglish-Swift model, containing approximately 72.6 million parameters, achieved a WER of 44.3% and CER of 29.6%, compared with 50.0% WER and 42.3% CER for zero-shot Whisper Small, which contains approximately 244 million parameters. The Hinglish Swift model also achieved an RTF of 0.41, compared with 4.31 for Whisper Small, making it approximately ten times faster in this experiment. These results indicate that domain-specific fine-tuning can improve both recognition accuracy and computational efficiency for Hindi-English code-switched speech, despite the fine-tuned model being substantially smaller.

The system additionally produces token-level language labels, language distributions, switch counts, switch-point sequences, and time-scaled code-switching timelines. Evaluation outputs can include WER and CER alignment information for categorizing recognition errors, along with language identification and code-switch detection metrics when appropriate ground-truth annotations are available. The application follows a measurement-honest approach: accuracy metrics are reported only when ground-truth data is available, approximate language-identification methods are explicitly labelled, and unconfigured models are shown as unavailable rather than being assigned placeholder values. These outputs provide both a practical demonstration of the implemented system and a reproducible basis for further experimentation, including fine-tuning and extension to additional Indian language pairs.

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